

T&M EQUIPMENT

Why Is Regular Calibration Important?

Calibration defines the accuracy and quality of your measurement. But just buying a certified, calibrated equipment is not the answer; you need to service and maintain the calibration of your equipment throughout its lifetime for reliable, accurate and repeatable measurements



Equipment calibration (Courtesy: Fluke Corporation)

■ SHWETA DHADIWAL BAID

Calibration defines the quality and accuracy of your measurement. It is the process of parameter verification of any equipment against a known standard so that the quantities around the world show

similar results. "Accurate measurements play a vital role at each stage in development and manufacturing of a quality product for consistent and reliable performance," says Anirudha Kela, managing director, Scientific Mes Techniq. A reliable and accurate result brings confidence in you as well as in your product.

The goal of calibration is to minimise the measurement uncertainty by ensuring the accuracy of your test equipment. S.M. Sulfikaruddin, vice president, support solutions unit, EMC, Agilent Technologies, says, "All the measurement processes are accompanied by errors and uncertainties, and they cannot be eliminated. Calibration quantifies and controls the errors or uncertainties of the measurement processes to an acceptable level."

Repeatability is a great virtue of equipment even ahead of accuracy, which can be defined only by quantification.

"In measuring, we quantify a thing, which firms up our knowledge, while qualifying a thing is never precise, like something is very good, too small or very big. When we say the voltage is 1.203V, it is precise, and it is nei-

ther less than 1.202V, nor more than 1.204V. The moment an equipment is calibrated by a standard that is the best earmarked, accuracy hierarchy is set up, irrespective of its origin, cost and newness. We therefore say we pass on the accuracy of a million-rupee meter to this thousand-rupee meter with one stroke of calibration that may cost just a few hundred rupees," explains S. Desai, managing director, Kalashri Electronics.

There are accreditation bodies around the world that have been evaluated by peers as competent to enhance the acceptance of products and services across national borders. This is to create an international framework to support international trade through the removal of technical barriers.

"The momentum is huge amongst world's leading test agencies regarding calibration. Everyone is demanding ISO/IEC 17025 accreditation including measurement uncertainty data," shares David Bisenius, an entrepreneur and internationally recognised regulatory expert in product safety.

Calibration is not a one-time thing. It's a repeatable process to maintain the accuracies. The interval for recalibrating your equipment is influenced by a lot of factors including the type, usage and environmental factors.

Traceability in calibration

Calibration is an unbroken chain of comparisons with national and international superior equipment. There are about 123+ countries bound to ISO